Université de Haute-Alsace
École Nationale Supérieure d'Ingénieurs
Sud Alsace

Informatique et Réseaux - 1A

Algorithm and Data Structures

TD1 : Introduction to Algorithms

Exercise 1

— Fibonacci

1. Implement the naive algorithm for calculating the nth term of the Fibonacci sequence.
2. Implement the dynamic programming algorithm for calculating such term.
3. Compare the runtime of each algorithm in **seconds**, use `clock()`[1].

Exercise 2

— Prime Numbers

1. Develop a naive algorithm to count the primes less than an integer n, with a time complexity of $\mathcal{O}(n\sqrt{n})$.
2. Develop a more efficient algorithm. Hint: multiples of prime numbers are not prime.
3. Show that the time complexity of the previously developed algorithm of question 2. is $\mathcal{O}(n\log(\log(n))$. Hint: Use the fact that:

$$\sum_{p\ prime\ ,\ p\leq n} \frac{1}{p} \leq \log(\log(n+1)) + M \tag{1}$$

   where M is the is the MeisselMertens constant [2].
4. Implement both naive and efficient algorithms and compare their runtime in seconds.

[1] Documentation C: `https://devdocs.io/c/chrono/clock`
[2] Meissel-Mertens constant: `https://fr.wikipedia.org/wiki/Constante_de_Meissel-Mertens`